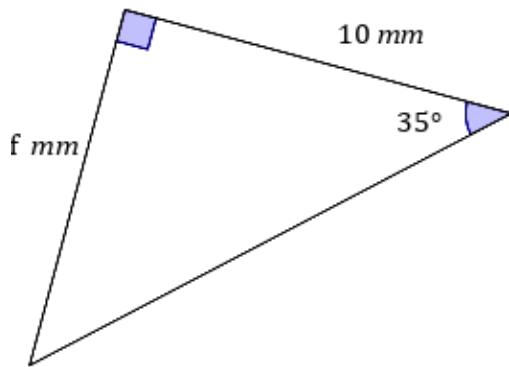


Solving for sides in right triangles



1 Consider the diagram shown.

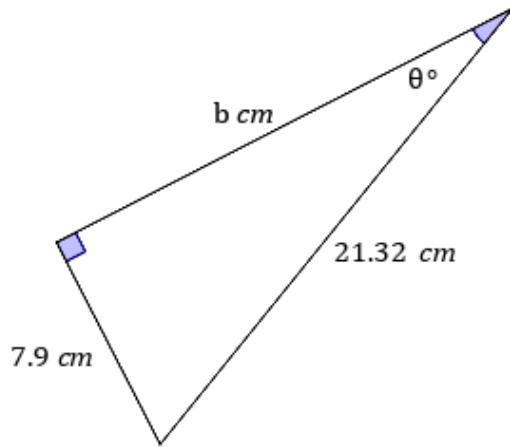


- a Is side with length f mm adjacent or opposite the 35° angle?
- b Find the length of the side adjacent to the 35° angle.
- c Should the length of the side opposite to the 35° angle be greater than or less than 10 mm?
- d Determine whether the following equations involving f are true.
 - i $f = \frac{\tan 35^\circ}{10}$
 - ii $f = \frac{10}{\tan 35^\circ}$
 - iii $\tan 35^\circ = \frac{10}{f}$
 - iv $\tan 35^\circ = \frac{f}{10}$
 - v $f = 10 \tan 35^\circ$
- e Hence, solve for f , correct to one decimal place.

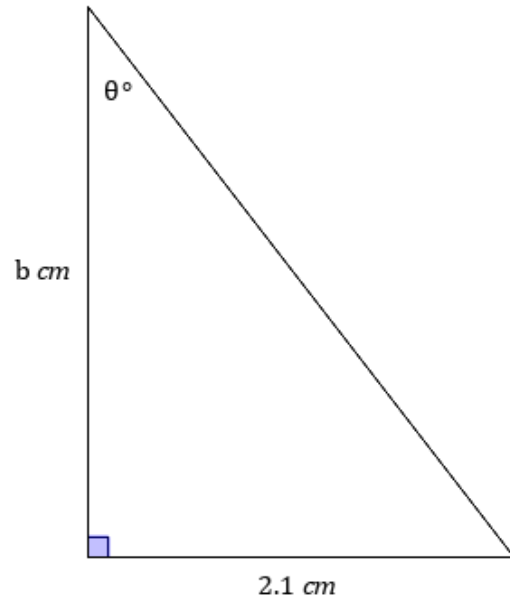
2 Given the value of $\tan \theta$, find the value of b in the following diagrams. Round your answers to one decimal place when necessary.



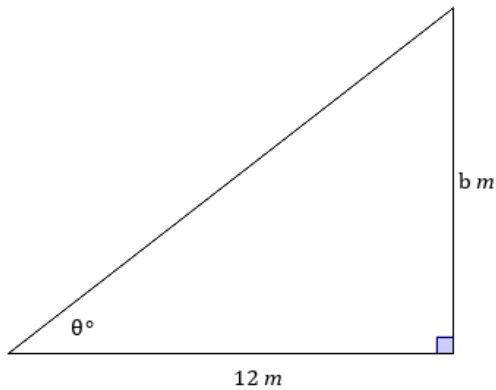
a $\tan \theta = 0.4$



b $\tan \theta = 0.3$

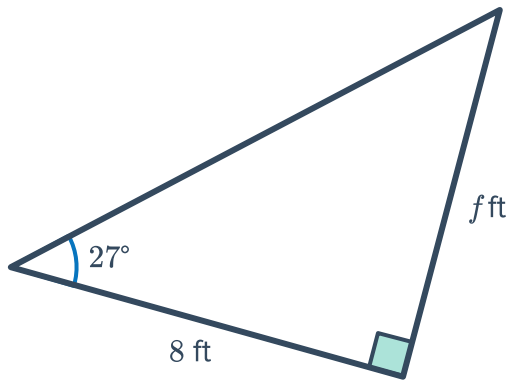


c $\tan \theta = \frac{2}{3}$

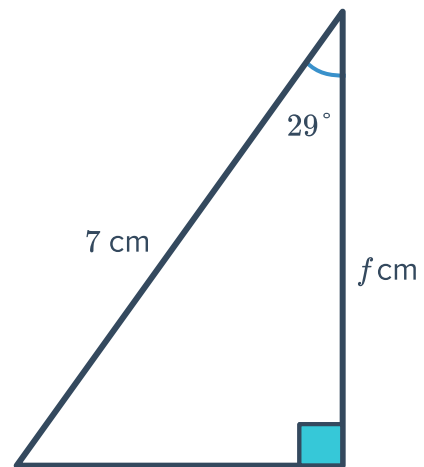


3 Find the value of the variables in the following right angles, rounding your answer to two decimal places when necessary.

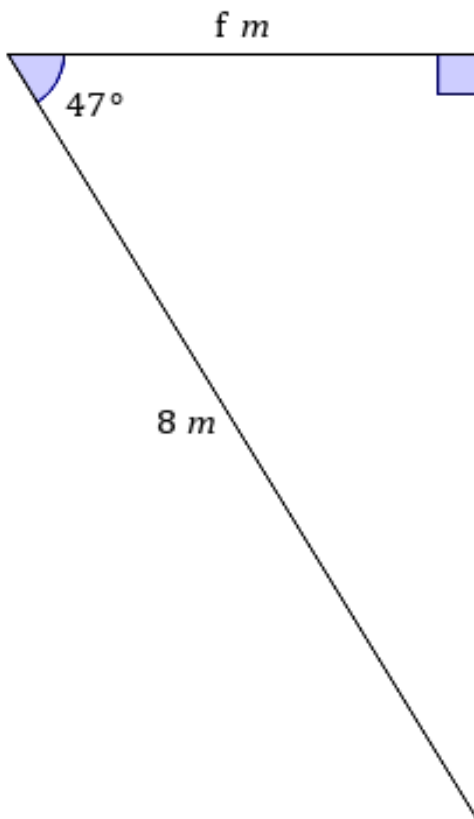
a



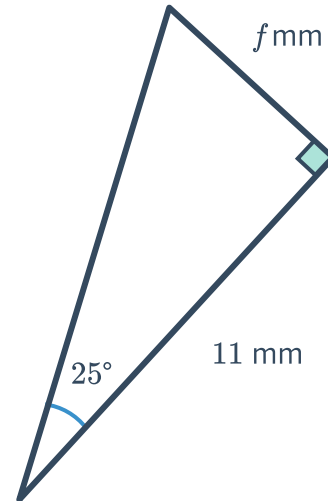
b



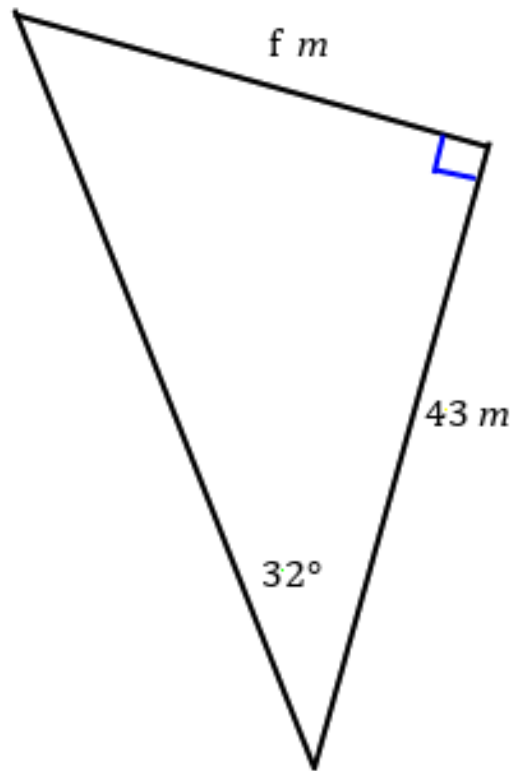
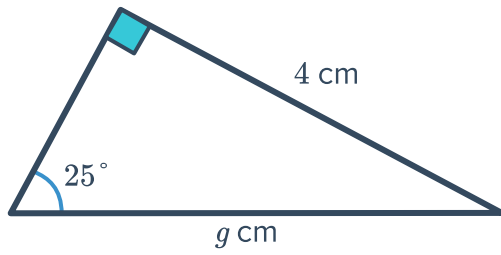
c



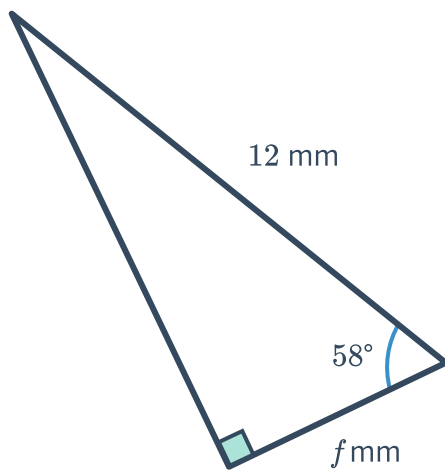
d



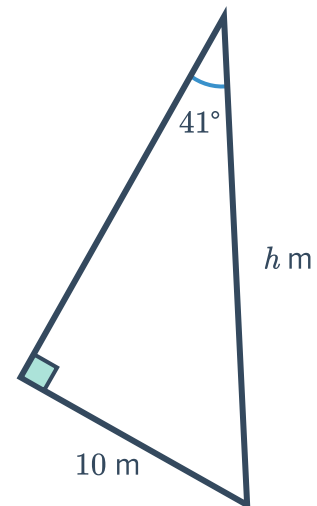
e



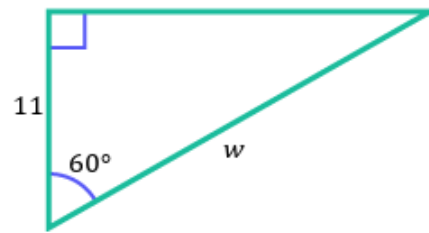
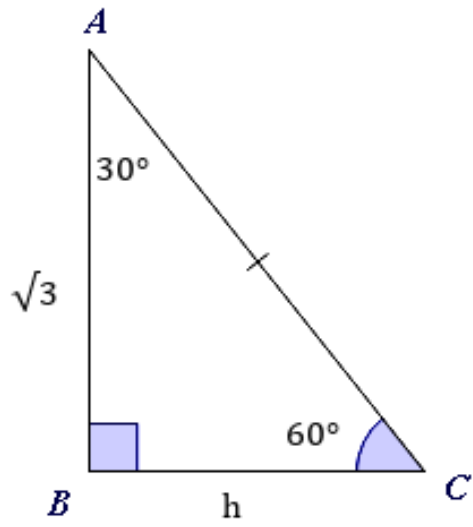
g



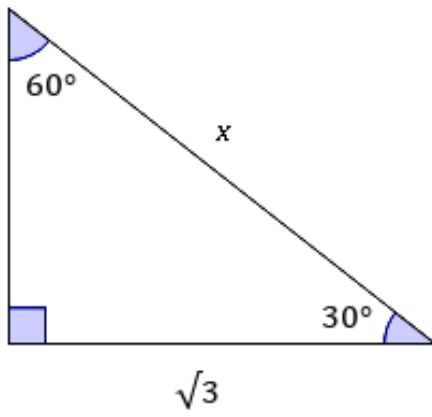
h



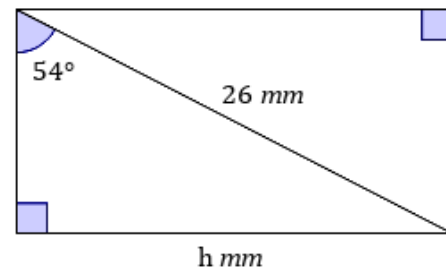
i



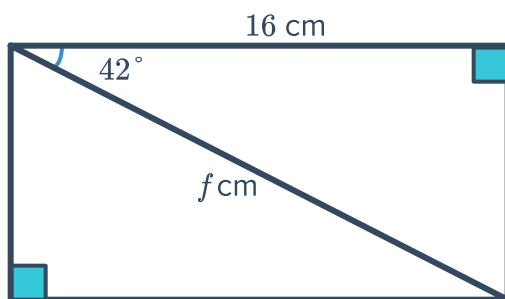
k



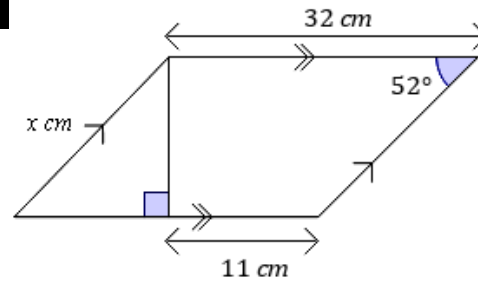
l



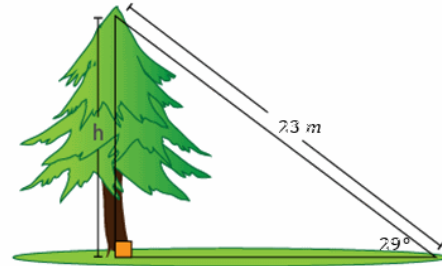
m



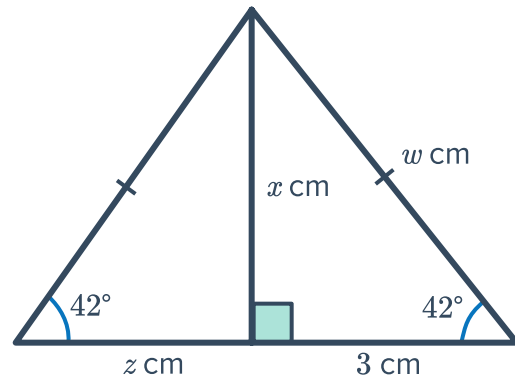
- 4 Consider the given diagram.
Find the value of x , the side length of the parallelogram. Round your answer to the nearest centimeter.



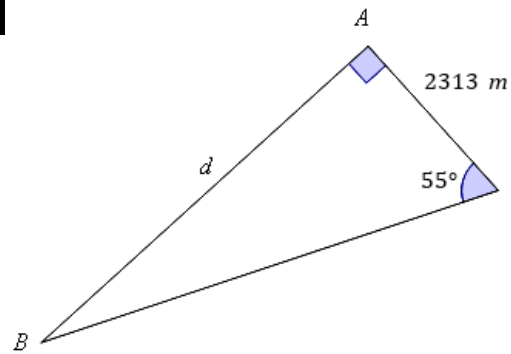
- 5 Consider the following figure.
If the height of the tree is h meter, find h to two decimal places.



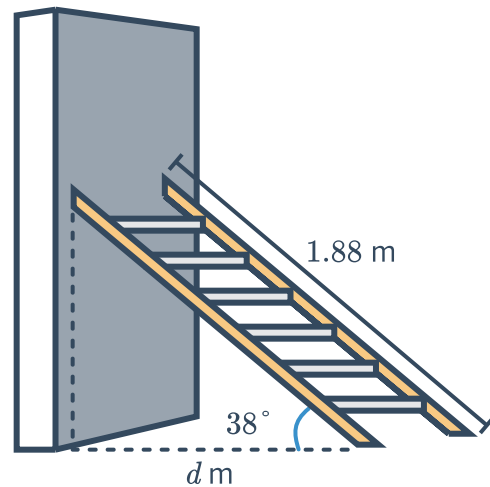
- 6 Consider the following diagram.
- Find the value of x , rounding your answer to two decimal places.
 - Find the value of w , rounding your answer to one decimal place.
 - Find the value of z , rounding your answer to the nearest whole number.



- 7 Neil's home is located at point A , and Neil is on his skateboard at point B .
If d is the distance between Neil and his home, calculate d , correct to two decimal places.

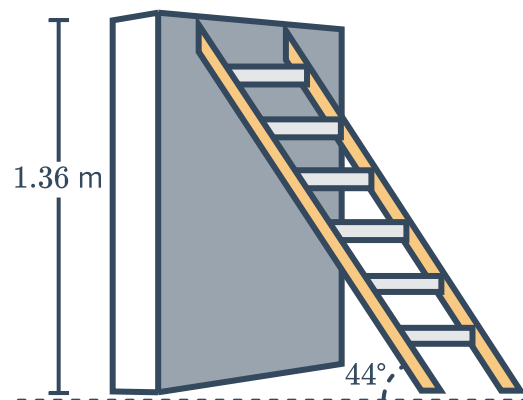


- 8 Consider the given figure.
If d is the distance between the base of the wall and the base of the ladder, find d to two decimal places.

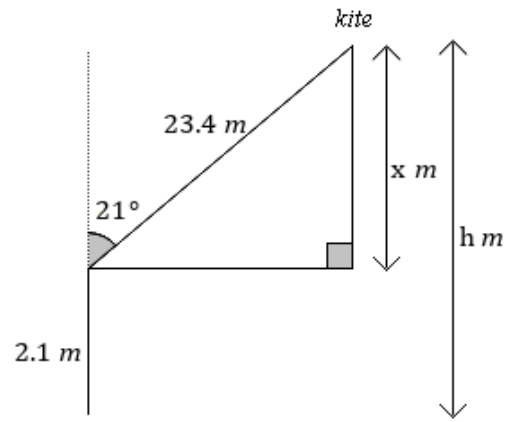


- 9 The longer side of a rectangular garden measures 14 m. A diagonal path makes an angle of 26° with the longer side of the garden.
If the length of the shorter side of the garden is y m, calculate y to two decimal places.

- 10 Consider the following figure.
If l is the length of the ladder in meters, find l , to two decimal places.

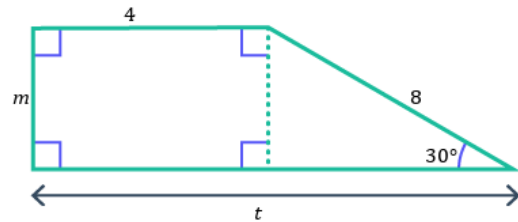


- 11 A girl is flying a kite that is attached to the end of a 23.4 m length of string. The angle between the string and the vertical is 21° . If the girl is holding the string 2.1 m above the ground:



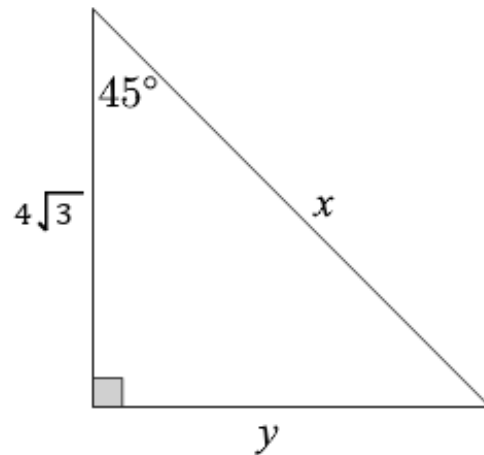
- Find x , correct to two decimal places.
- Hence find the height h of the kite above the ground, correct to two decimal places.

- 12 Consider the unknown lengths in the given diagram.



- Find exact value of m .
- Find exact value of t .

- 13 Consider the adjacent diagram.

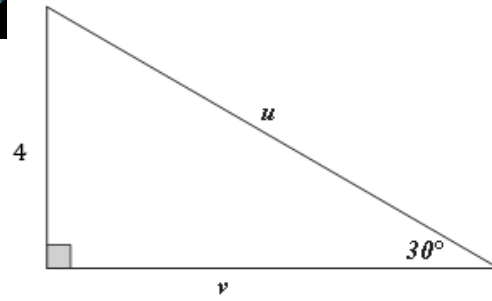


- Solve for the exact value of x .
- Solve for the exact value of y .

14 Consider the given diagram.

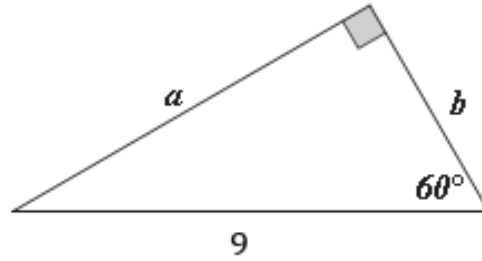


- a Solve for the exact value of v .
- b Solve for the exact value of u .



15 Consider the adjacent diagram.

- a Solve the exact value of a .
- b Solve the exact value of b .



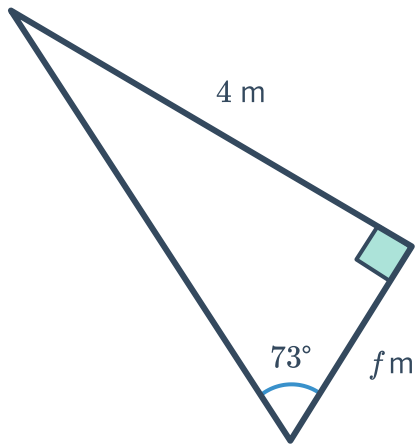
Additional questions

16 Evaluate the following, rounding your answer to two decimal places:

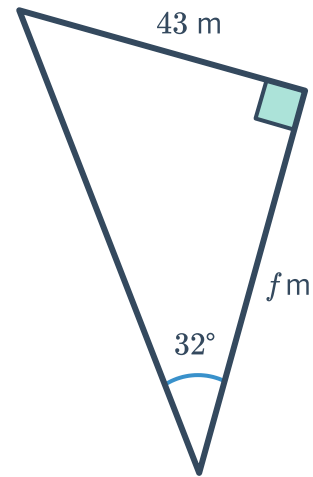
- | | | | |
|---------------------|---------------------|-----------------------------|----------------------------|
| a $\sin 82^\circ$ | b $\cos 28^\circ$ | c $\tan 84^\circ$ | d $9 \sin 63^\circ$ |
| e $7 \cos 77^\circ$ | f $8 \tan 28^\circ$ | g $\frac{\sin 71^\circ}{7}$ | h $\frac{\cos 7^\circ}{9}$ |

- 17 Find the value of the variable in the following triangles, rounding your answer to two decimal places:

a

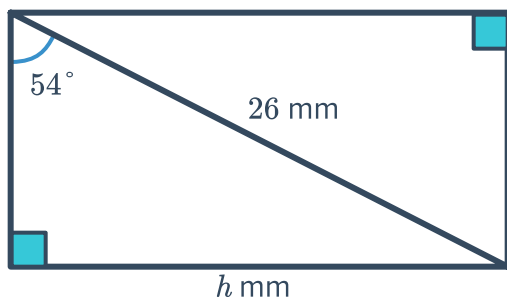


b

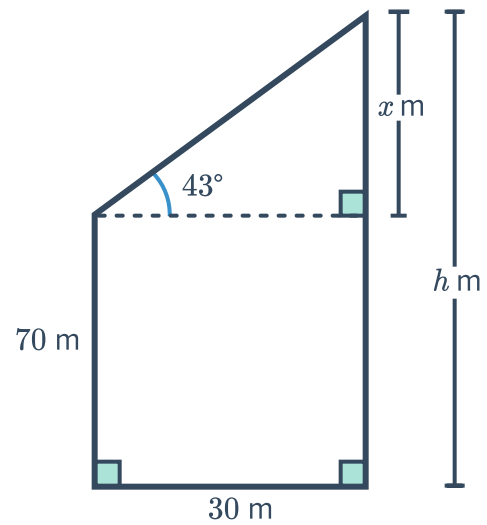


- 18 Find the value of the variable(s) in the following diagrams, rounding your answer to two decimal places:

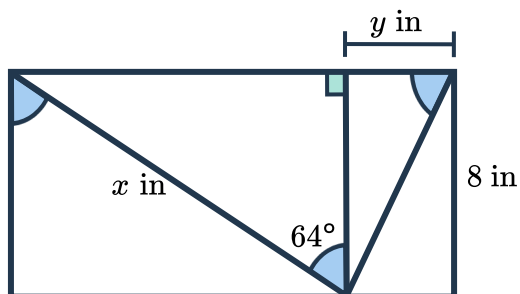
a



b



c





19 Consider the shape given.

- a Find the value of d , rounding your answer to one decimal place.
- b Find the value of f , rounding your answer to one decimal place.

